

21) Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$

Does $\{v_1, v_2, v_3\}$ span $\mathbb{R}^4$? Why or why not?

Solution: Let $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$ be an arbitrary vector in $\mathbb{R}^4$.
By defn. of $\text{span}\{v_1, v_2, v_3\}$ span $\mathbb{R}^4$ if the vector $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$ is a linear combination of the vectors v_1, v_2 and v_3 , i.e.

$$x_1 v_1 + x_2 v_2 + x_3 v_3 = b \text{ for some } x_1, x_2, x_3 \in \mathbb{R}.$$

From Theorem 4, this is equivalent to claiming that the matrix equation $Ax = b$ has a solution for all b , where $A = [v_1 \ v_2 \ v_3]$.

From Theorem 4, this is equivalent to claiming that the matrix A has a pivot position in every row.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ + \text{Row 1}}]{\substack{\text{Row 1} = \text{Row 1} \\ - \text{Row 3}}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ \times (-1)}]{\substack{\text{Row 4} = \text{Row 4} \\ + \text{Row 2}}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix} \xrightarrow[\text{Row 4} = \text{Row 4} + \text{Row 3}]{\text{Row 4} = \text{Row 4} + \text{Row 3}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in reduced row echelon form.

1. Row 1, 2 and 3 are non-zero rows. Row 4 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.

2. Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 1 in Column 2
Row 3 Leading Entry = 1 in Column 3

$\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.